and frequency for measuring volatility. This introduces data-snooping bias. Second, volatility and correlation are inherently unstable and nonstationary. Their use therefore introduces additional estimation risk and potential portfolio instability. We take a simpler approach that accomplishes much the same thing as traditional risk parity. Starting with the MSCI US and long Treasury bond indexes used in our 60/40 portfolio, we add REITs, credit bonds, and gold, with an equal weighting given to each asset class.[6] We use credit bonds to increase the fixed-income exposure of the portfolio. Credit bonds diversify our fixed-income allocation by providing some credit risk premium with less duration risk than long Treasuries. REITs give us exposure to real assets with some additional risk exposure to equities. Gold gives us real asset exposure that is different from real estate.[7] Gold has the highest volatility, and so it represents only 20% of our parity portfolio, whereas bonds receive the largest allocation of 40%, being represented twice in the portfolio. Exposure to equities is somewhere between gold and bonds.

By structuring our portfolio purposefully to begin with, we are able to balance our risk exposure between fixed income, equities, and real assets nonparametrically without incurring any added estimation risk. We will see that the addition of absolute momentum to our parity portfolio reduces and equalizes risk exposure across all asset classes.

Table B.5 shows the correlations of the S&P 500, U.S. 10 Year Treasury, and GSCI Commodity indexes to the 60/40 and parity portfolios, both with and without 12-month absolute momentum. Our parity portfolio with 12-month absolute momentum shows a modest and nearly equal correlation to both stocks and bonds. Because of the downside risk attenuation through absolute momentum, we have achieved risk parity while limiting fixed-income assets to no more than 40% of our portfolio.

Table B.5 Monthly Correlations, 1974–2012

	60/40 Portfolio	60/40 w/Abs Momentum	Parity Portfolio	Parity w/Abs Momentum
S&P 500	0.92	0.67	0.67	0.40
10-yr Treasury	0.58	0.35	0.37	0.36
GSCI	0.05	0.06	0.25	0.19